

Metallography

Aim :

To polish a mounted mild steel specimen using sandpapers of different grades for metallographic examination.

Procedure :

1. A mounted mild steel specimen was given for the experiment.
2. The surface of the specimen was polished using sandpapers of different grades, starting from the coarsest grade and gradually moving towards finer grades in the following order:
$$400 \rightarrow 600 \rightarrow 800 \rightarrow 1000 \rightarrow 1200 \rightarrow 1500 \rightarrow 2000 \rightarrow 3000 \leftarrow 2500$$
3. The specimen was first polished using 400-grade sandpaper to remove the roughness and major surface scratches.
4. The sandpaper was changed successively to finer grades because the coarser sandpaper produces relatively deep scratches, while finer grades gradually remove these scratches and produce a smoother surface.

5. Whenever the sandpaper was changed to the next grade, the specimen was rotated by 90° . This was done so that the scratches produced by the new sandpaper were perpendicular to those produced by the previous grade.
6. The 90° rotation helps in identifying the scratches from the previous grade. Polishing with the new grade was continued until the scratches in the previous direction were no longer visible.
7. The same procedure was repeated successively with 600, 800, 1000, 1200, 1500, 2000, 2500, 3000-grade sandpapers.
8. Finally, the specimen obtained a smooth and uniformly surface, making it suitable for further metallographic examination.